

System and Network Security

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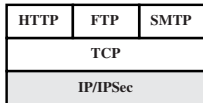
Transport-Level Security

Comparison of Threats on the Web

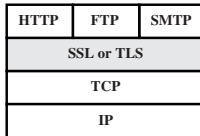
	Threats	Consequences	Countermeasures
Integrity	• Modification of user data • Trojan horse browser • Modification of memory • Modification of message traffic in transit	• Loss of information • Compromise of machine • Vulnerability to all other threats	Cryptographic checksums
Confidentiality	• Eavesdropping on the net • Theft of info from server • Theft of data from client • Info about network configuration • Info about which client talks to server	• Loss of information • Loss of privacy	Encryption, Web proxies
Denial of Service	• Killing of user threads • Flooding machine with bogus requests • Filling up disk or memory • Isolating machine by DNS attacks	• Disruptive • Annoying • Prevent user from getting work done	Difficult to prevent
Authentication	• Impersonation of legitimate users • Data forgery	• Misrepresentation of user • Belief that false information is valid	Cryptographic techniques

- One way to provide Web security is to use IP security (IPsec). The advantage of using IPsec is that it is transparent to end users and applications and provides a general-purpose solution. Furthermore, IPsec includes a filtering capability so that only selected traffic need incur the overhead of IPsec processing.
- Another relatively general-purpose solution is to implement security just above TCP. The foremost example of this approach is the Secure Sockets Layer (SSL) and the follow-on Internet standard known as Transport Layer Security (TLS). At this level, there are two implementation choices. For full generality, SSL (or TLS) could be provided as part of the underlying protocol suite and therefore be transparent to applications. Alternatively, SSL can be embedded in specific packages. For example, Netscape and Microsoft Explorer browsers come equipped with SSL, and most Web servers have implemented the protocol.
- Application-specific security services are embedded within the particular application (for example, PGP or S/MIME, Kerberos and X.509 authentication services, Password Management)

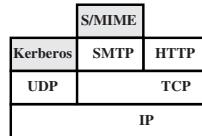
Relative Location of Security Facilities in the TCP/IP Protocol Stack



(a) Network level



(b) Transport level



(c) Application level

Secure Socket Layer (SSL)/Transport Layer Service (TLS)

- **Secure Socket Layer (SSL)** provides security services between TCP and applications that use TCP.
- The Internet standard version is called **Transport Layer Service (TLS)**.
- SSL/TLS provides confidentiality using symmetric encryption and message integrity using a message authentication code.
- SSL/TLS includes protocol mechanisms to enable two TCP users to determine the security mechanisms and services they will use.

- **HTTPS (HTTP over SSL)** refers to the combination of HTTP and SSL to implement secure communication between a Web browser and a Web server.
- The HTTPS capability is built into all modern Web browsers. Its use depends on the Web server supporting HTTPS communication. For example, search engines do not support HTTPS.
- The principal difference seen by a user of a Web browser is that URL (uniform resource locator) addresses begin with `https://` rather than `http://`.
- A normal HTTP connection uses port 80. If HTTPS is specified, port 443 is used, which invokes SSL.

When HTTPS is used, the following elements of the communication are encrypted:

- URL of the requested document
 - Contents of the document
 - Contents of browser forms (filled in by browser user)
 - Cookies sent from browser to server and from server to browser
 - Contents of HTTP header
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- HTTPS is documented in RFC 2818, HTTP Over TLS.
 - There is no fundamental change in using HTTP over either SSL or TLS, and both implementations are referred to as HTTPS.